Final Project Report: Wine Quality Prediction

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Abstract:

This report evaluates how accurately wine quality can be predicted using chemical properties from the UCI Wine Quality dataset. We analyzed separate datasets for white and red wine using linear regression, regression trees, and random forests to evaluate the predictive performance of each of these models. Model results were evaluated mainly through in-sample and out-of-sample mean squared error, along with visual diagnostics such as residual plots, tree outputs, random forest error plots, and more. The findings suggest that random forests produced the strongest predictions for both wine types, showing that chemical measurements can help predict wine quality but flexible models are better suited for capturing patterns in the data than simpler linear approaches.

Introduction:

This project analyzed whether physicochemical properties of wine can be used to predict wine quality scores. Our team worked with the UCI Wine Quality dataset, which includes separate red and white wine datasets, and compared three predictive methods: linear regression, regression trees, and random forests. Throughout the project, we completed exploratory data analysis, built benchmark and selected regression models, evaluated tree-based models, and compared model accuracy using in-sample and out-of-sample MSE. Overall, our results showed that random forests performed the best for both red and white wine because they produced the lowest prediction errors and were better able to capture complex relationships in the data. Alcohol was also consistently identified as one of the most important predictors of wine quality across the models and variable importance plots.

Project Background and Goal:

We used the Wine Quality dataset from the UCI Machine Learning Repository to predict wine quality using measurable chemical properties from physicochemical lab tests rather than factors such as brand, grape variety, or price. The main goal of this project is to measure how strongly these chemical variables relate to quality and how accurately they can predict quality scores. To do this, our study compares multiple modeling approaches, including linear regression and more flexible methods such as random forests, to determine which method best captures the relationship between the input variables and wine quality in a predictive and interpretable way. Another goal is to determine which modeling approach has the best predictive accuracy.

Data Description:

The dataset comprises two related files concerning red and white Portuguese "Vinho Verde" wines from northern Portugal.There are two separate datasets for white and red wine. The white wine dataset consists of 4,898 observations, while the red wine dataset consists of 1,599 observations. Both have 11 input features, and 1 output variable, quality. Predictors include fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. Without missing values, the dataset can be treated as a regression problem or a classification problem. This project will treat it as a regression problem due to the numerical quality scores as the response variable. Here the response variable is quality measured by a score from 0 to 10 indicating sensory appraisal. This variable is numeric and ordered, therefore, it suits being represented with regression techniques since a qualitative assessment on the quality of wine can determine how wine quality changes when different chemical components of wine are present. The hope is to employ these physicochemical indicators as predictors of this quality index and to see if the regression model can give any value to predictive power. Many predictor variables are quite clear in practical application in wine chemistry. For example, fixed acidity is the non-volatile acids present in wine, and volatile acidity is mostly associated with acetic acid, which when too high can have a negative impact on flavor. Additionally, residual sugar represents the remaining sugar after fermentation, and chlorides a measure of salt content. As a whole, these variables give a quantitative view of each sample's kind and form the basis for making quality predictions.

This makes the dataset useful for testing whether measurable chemical properties can help explain differences in wine quality. However, the dataset does not include other factors that may affect quality ratings, such as grape type, brand, price, or region. Because of this, the models only evaluate wine quality based on the chemical measurements provided in the data.

Preliminary Analysis

After conducting exploratory data analysis, we have discovered clear differences between the white and red wine datasets. From the summary statistics, white wine had higher average residual sugar (6.39 vs. 2.54) and total sulfur dioxide (138.4 vs. 46.47), while red wine had higher fixed acidity (8.32 vs. 6.86) and volatile acidity (0.53 vs. 0.28). However, both of the wines had very similar quality scores, with white wine scoring at 5.88, and red wine scoring 5.64. The histograms confirm this by revealing a clear concentration of wines in the mid range, indicating most the dataset contains average-quality wines (Figure 1). White wine scores range from 3 to 9, while red wine scores range from 3 to 8. This may affect how well the regression model predicts quality scores at the lower and upper extremes, indicating the need for a model that can handle outliers well.

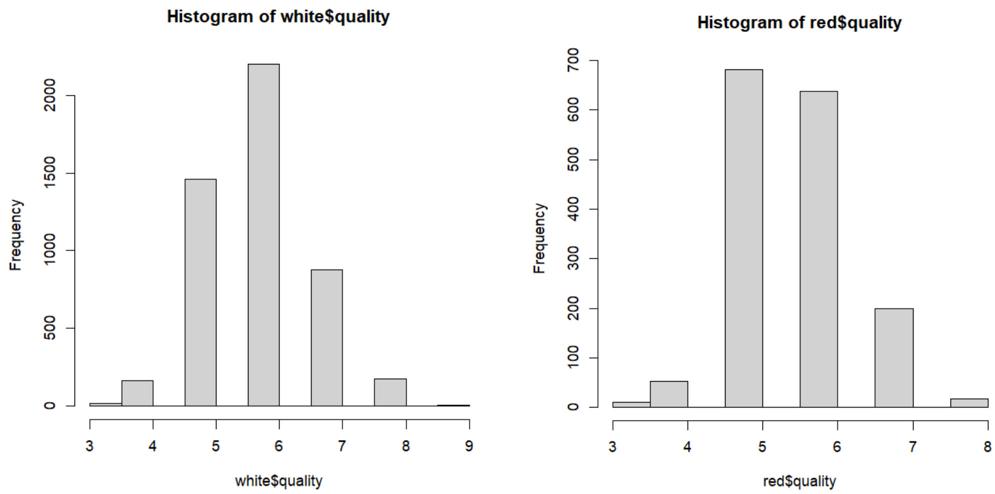


Figure 1. White and Red Wine Quality Histograms

Additionally, the white wine correlation heatmap (Figure 2) reveals that alcohol has the strongest positive correlation (+0.6) with quality, and density has the lowest correlation (-0.4) with quality. This suggests that white wines with higher alcohol content and lower density are more likely to receive higher scores. Similarly, red wine heatmap (Figure 2), reveals strong positive correlation with alcohol and the lowest correlation with volatile acidity (-0.6).

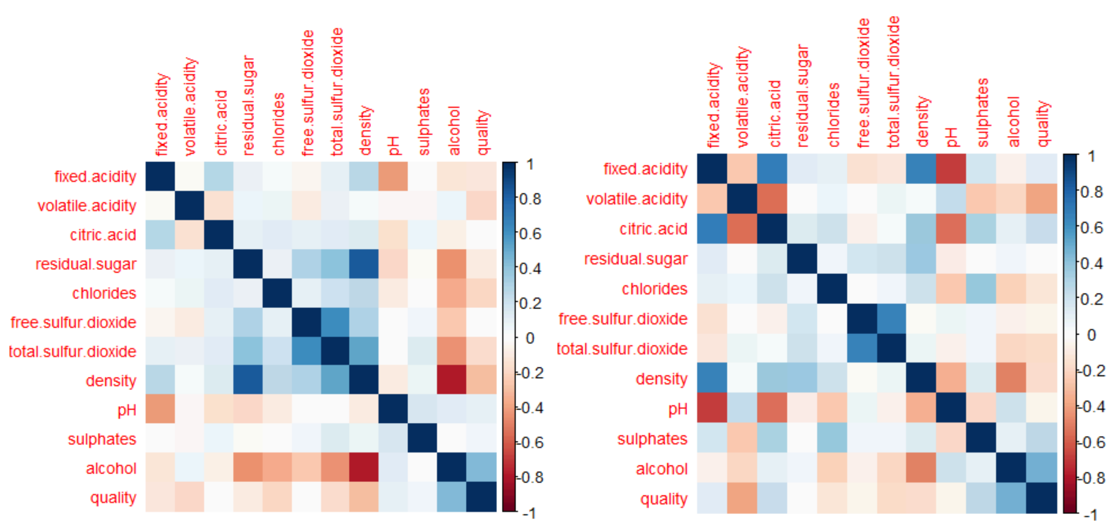


Figure 2. White and Red wine correlation map

To further prove the relationship, we fit a benchmark linear regression model for both wine and red wine using all 11 physiochemical predictors and quality as the response variable. For white wine, the model is considered statistically significant, with an adjusted R squared of 0.2985, indicating that predictor variables explain around 29% of the variation in white wine quality. The significant predictors at the 5% level include volatile acidity, residual sugar, free sulfur dioxide, density, pH, sulphates, alcohol, and fixed acidity. For red wine, the model is also statistically significant, with a R squared of 0.3479. The significant predictors for red wine include volatile acidity, chlorides, total sulfur dioxide, sulphates, and alcohol. The chart below summarizes the benchmark model's results:

	White wine	Red wine
MSE (in sample)	0.6103062	0.4235515
MSE (out of sample)	0.5940786	0.390562
Adjusted R squared	0.2984545	0.347958
AIC	3387.084	2861.438
BIC	3455.617	2929.971

Figure 3. Benchmark Results

Overall, the preliminary analysis confirms that the wine quality dataset is feasible for regression modeling since many predictors show statistically significant relationships with quality. However, both wines have relatively low R squared values which suggest that a simple linear regression model might not be enough to fully capture their relationships and might need more complicated modeling approaches.

Method 1 - Linear Regression:

For our first prediction method, we decided to use linear regression because wine quality is measured as a numerical score from 0 to 10. After fitting the benchmark models with all the psychicochemical predictors, best subsets variable selection with BIC was used to create simpler models (Figure 4). For white wine, the selected model used 7 predictors: fixed acidity, volatile acidity, residual sugar, free sulfur dioxide, density, pH, and sulphates. This model had an in-sample MSE of 0.6101, a 5-fold cross-validation MSE of 0.5741, and an adjusted R-squared of 0.2987. For red wine, the selected model used 6 predictors: volatile acidity, chlorides, total sulfur dioxide, pH, sulphates, and alcohol. This model had an in-sample MSE of 0.4239, a 5-fold cross-validation MSE of 0.4243, and an adjusted R-squared of 0.3474. Overall we said that the red wine linear regression model performed better because it had lower prediction error and a higher adjusted R-squared than the white wine model. Figure 5 summarizes the best subsets model results.

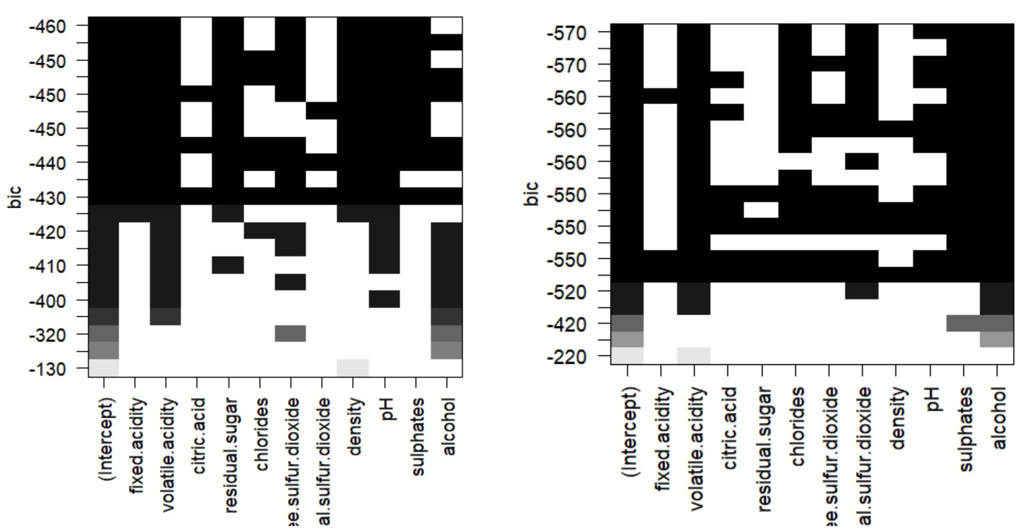


Figure 4. BIC Best Subsets Variable Selection

	White Wine	Red Wine
MSE	0.6101134	0.423912

Adjusted R squared	0.298676	0.3474029
AIC	3382.658	2857.696
BIC	3430.103	2899.87
5 fold Cross Validation MSE	0.5740815	0.4242664

Figure 5. Best Subsets Results

The residual diagnostic plots showed that linear regression was useful but also confirmed not the best fit for data. In the residuals versus fitted plots, the residuals were not randomly scattered around zero and showed visible patterns, which suggests that the relationship between the chemical predictors and wine quality may not be fully linear (Figure 5). The Q-Q plots also showed that the residuals did not fall closely along the reference line, especially for white wine, where the tail suggested possible extreme residuals or outliers (Figure 6).This suggests that the normality assumption was probably not met. Therefore, linear regression served as a helpful benchmark model, but the residual plots suggest that more flexible methods, such as regression trees and random forests, may better capture nonlinear patterns in the wine quality data.

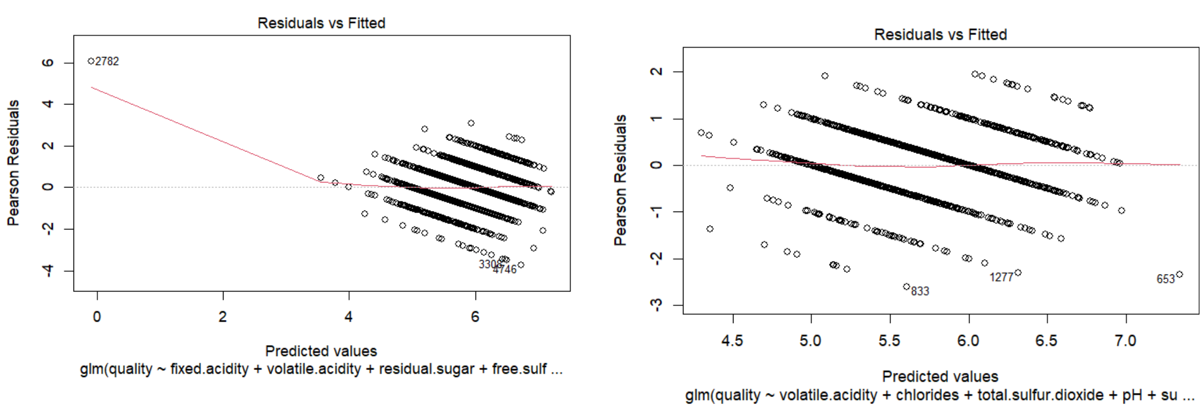


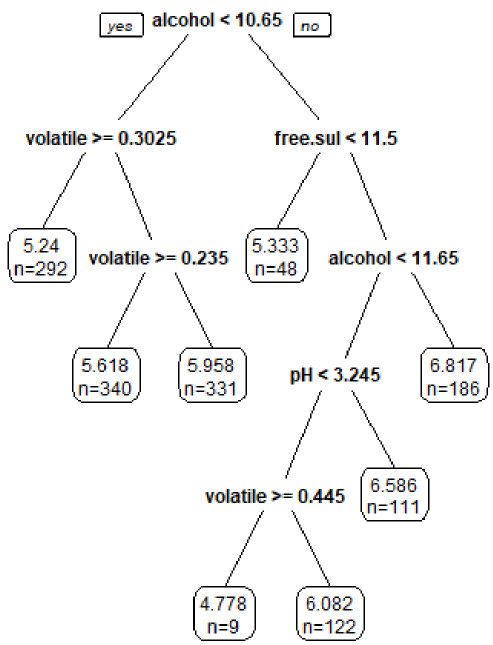
Figure 6. Residuals vs Fitted Plots for White (left) and Red (right) Wine

Figure 7. Q-Q Residuals for White (left) and Red (right) Wine

Method 2 - Regression Tree:

We used regression trees for the second prediction method to model wine quality in a more flexible way than linear regression. Unlike linear regression, regression trees do not just assume that the relationship between the predictors and quality is linear, but rather splits the data into smaller groups based on the predictor values, giving each group a predicted quality score. For the white wine dataset, the regression tree produced an in-sample MSE of 0.5956 and an out-of-sample MSPE of 0.5743. The tree showed that alcohol was one of the most important splitting variables, as it used alcohol as the first node to divide wines into different predicted quality scores. The highest predicted white wine quality score was 6.817, which occurred for wines with higher alcohol and moderate free sulfur dioxide levels. The lowest predicted quality score was 4.778, which was for wines with lower alcohol and higher volatile acidity (Figure 8).

For the red wine dataset, our results showed that the regression tree performed slightly better, with an in-sample MSE of 0.4001 and an out-of-sample MSPE of 0.4175, which is lower than the white wine tree. Red wines also had higher quality scores compared to white wine, with the highest predicted quality of 6.857, and lowest score of 4 (Figure 9). Regression trees captured nonlinear patterns that linear regression missed, making them an helpful intermediate model as they improved interpretability and flexibility. However, they were not the strongest overall method when compared to random forests. Figure 10 shows a summary chart of the tree results.



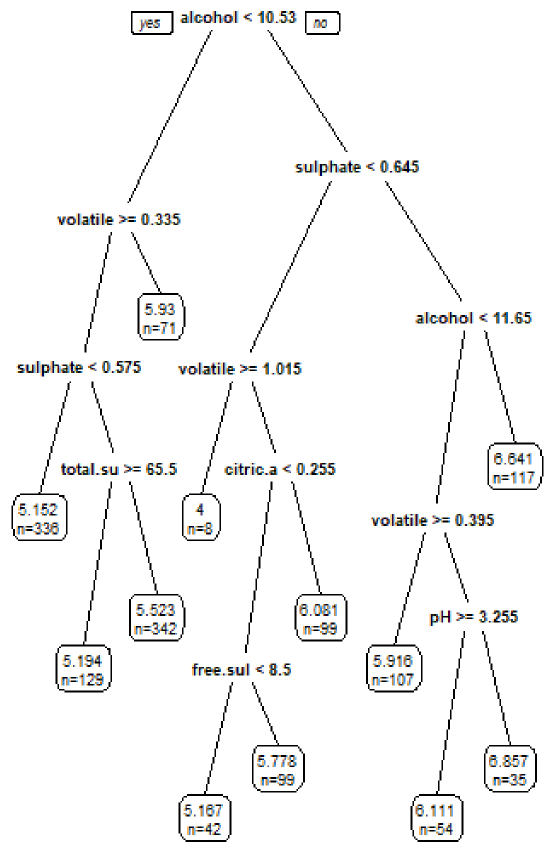


Figure 8. White Wine Regression Tree

Figure 9. Red Wine Regression Tree

	White Wine	Red Wine
MSE (in sample)	0.5955841	0.4005961
MSPE (out of sample)	0.574334	0.4174576

Figure 10. Regression Tree results

Method 3 - Random Forests:

For the final method, we used random forests, a more advanced method that builds many regression trees and combines their results. Instead of relying on a single tree, random forests run multiple trees using different samples of the data and then aggregate their outputs to help get us results. This approach improves prediction accuracy and lowers the chance of overfitting which is helpful since our data has outliers. In the random forest error plots, the error dropped quickly at first as more trees were added, and then it leveled off as the model got closer to 500 trees (Figure 11). This indicates that increasing the number of trees improved the model early on, but the prediction error eventually stabilized. For the white wine dataset, the random forest model produced an out-of-sample MSE of 0.5337, which was lower than the linear regression and regression tree models. For the red wine dataset, the random forest model produced an out-of-sample MSE of 0.2774, which was the lowest error among all three methods. Random forest results are summarized below in figure 13.

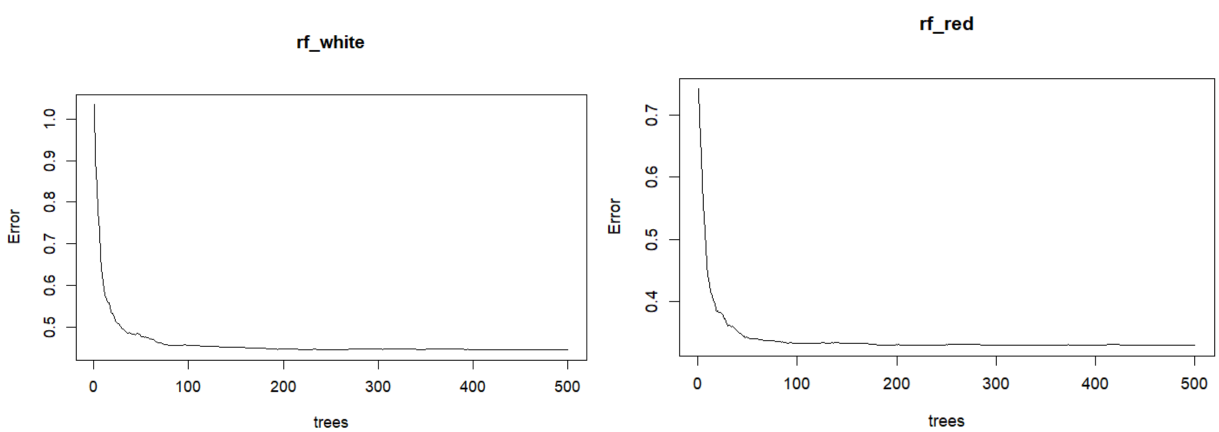


Figure 11. White and Red Wine Random Forest Plots

Looking at the variable importance plots tells us which predictors played the largest role in predicting wine quality. For both wine types, alcohol stood out as the most important factor in predicting quality, confirming our preliminary analysis. In white wine, density and free sulfur dioxide had moderate importance in the model, while in the red wine, sulphates and volatile acidity had high importance after alcohol (Figure 12). In conclusion, the random forest model performed the best because it had the lowest prediction error and was better able to capture complex, nonlinear relationships between the physicochemical predictors and wine quality scores. Its main limitation is that it is less directly interpretable than a single regression tree because it combines many trees instead of showing one simple decision path.

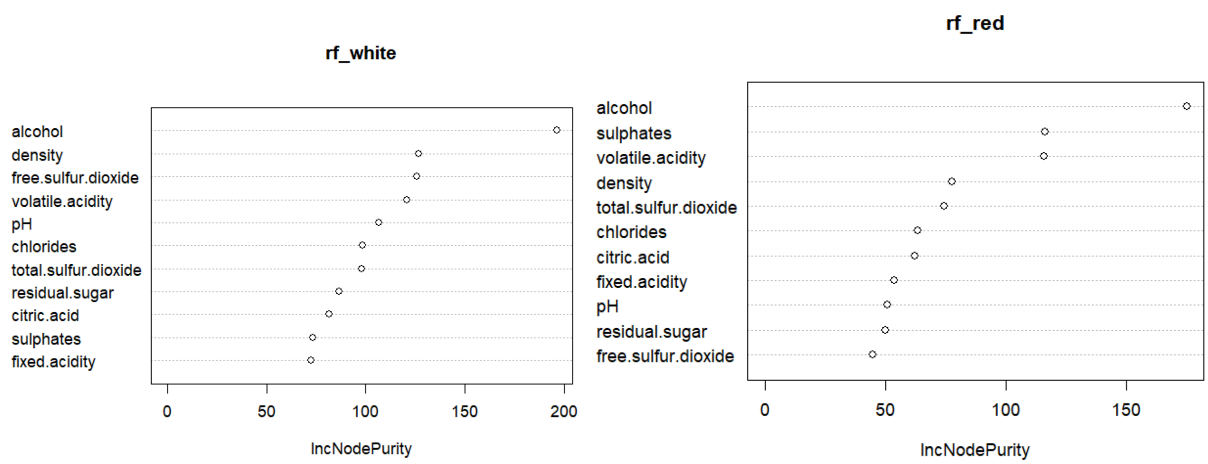


Figure 12. White and Red Wine Variable Importance Plots

	White wine	Red wine
MSE (in sample)	0.4439656	0.3307936
MSE (out of sample)	0.5336994	0.2774283

Figure 13. Random Forest Results

Results Comparison:

Overall, random forests performed the best for both wine types, as it produced the lowest out-of-sample MSE. For white wine (Figure 14), random forest had an out-of-sample MSE of 0.5337, which was lower than best subsets linear regression (0.5741) and the regression tree (0.5743). For red wine (Figure 15), random forest performed even better with an out-of-sample MSE of 0.2774, compared to best subsets linear regression (0.4243) and the regression tree (0.4175). This shows that random forest was better at capturing complex and nonlinear relationships between the predictors and quality scores. The figures further supported this conclusion because the random forest error plots showed the model's prediction error decreasing as more trees were added and then stabilizing, which suggests the model became more reliable. Linear regression was useful as a simple and interpretable benchmark, but the residual plots showed patterns and non-normality, suggesting that the linear model did not fully fit the data. The regression tree was way easier to interpret and showed important decision rules, such as alcohol being a key split, confirmed in the variable importance plots. However, because it was still less accurate than our final method, we selected random forest as the strongest overall model due to its strongest predictive accuracy in determining wine quality.

Method	In sample MSE	Out of sample MSE
Benchmark linear regression	0.6103	0.5941
Best subsets linear regression	0.6101	0.5741 (from cross validation)
Regression Tree	0.5956	0.5743
Random Forest	**<u>0.4439</u>**	**<u>0.5337</u>**

Figure 14. White Wine Results Comparison

Method	In sample MSE	Out of sample MSE
Benchmark linear regression	0.4236	0.3906
Best subsets linear regression	0.4239	0.4243 (from cross validation)
Regression Tree	0.4006	0.4175
Random Forest	**0.3308**	**0.2774**

Figure 15. Red Wine Results Comparison

Conclusion:

In this project, we have learned how to predict wine quality using different data mining methods using the chemical properties of red and white wine. By conducting linear regression, regression trees, and random forests, we were able to determine each modeling approach's predictive accuracy in predicting wine quality scores. Each method gave us different results and also showed different levels of accuracy, which allowed us to compare their performance.

Based on our findings, it appears that random forests performed the best out of all the models we tested sice it produced the lowest error values and most accurate predictions for both red and white wines. Regression trees performed better than linear regression and improved interpretability, but they were still less accurate than random forests. Linear regression had the weakest performance because it could not fully capture the complex relationships in the data. Overall, random forests were the best method for this dataset because they captured more complex relationships while producing the most accurate predictions of wine quality.

References:

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